An Adaptive Explainable Cognitive Graph Digital Twin for Cyber-Resilient Smart Hospital Energy Ecosystems

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Abstract. Smart hospital campuses rely on localized microgrids to sustain intensive-care ventilators, surgical robotics, and biomedical cold chains. A stealthy False Data Injection Attack (FDIA) on substation Remote Terminal Units can corrupt the energy-management state estimator while evading its χ^2 residual check, placing patient-critical loads at risk. We introduce a Cognitive Graph Digital Twin (CGDT) that mirrors the hospital electrical topology as a dynamic admittance graph and runs a multi-task Admittance-Conditioned GraphSAGE backbone for simultaneous attack detection and bus-level localization in a single forward pass at 0.908 ms latency. A novel Attribution-Guided Multi-Task Loss aligns Integrated-Gradients feature attributions with ground-truth anomaly labels during training, resolving the post-hoc explainability gap and lifting localization F1 by 5.52 percentage points. The digital twin maintains a formally defined state-space synchronization model with an exponentially weighted rolling baseline. Upon detecting an attack, a closed-loop healing module quarantines compromised sensor channels and reconstructs measurements via graph-guided spatial interpolation, reducing Weighted Least Squares estimation error by 99.97%. On the IEEE 14-bus test feeder under full nonlinear AC power flow, the CGDT achieves $98.25\% \pm 0.34\%$ detection F1 and $97.62\% \pm 0.84\%$ localization F1 across five random seeds, and retains 92.80% detection F1 under zero-shot topology reconfigurations without retraining.

Keywords: Graph Digital Twin · Healthcare Cybersecurity · GraphSAGE · Explainable AI · False Data Injection Attacks · Smart Hospital Microgrid

1 Introduction

The digital transformation of tertiary healthcare facilities has produced cyber-physical ecosystems where uninterrupted electrical supply is not merely an operational convenience but a patient-safety imperative. Modern smart hospitals

operate localized microgrids that integrate utility feeders, rooftop photovoltaic arrays, battery energy storage, and emergency diesel generators to power life-critical loads: mechanical ventilators and extracorporeal membrane oxygenation circuits in intensive-care units, robotic surgical suites requiring sub-second voltage stability, mRNA vaccine cold-chains maintained at $-80°$C, and Internet-of-Medical-Things (IoMT) edge gateways linking bedside monitors to electronic health record servers [5,19]. An undetected voltage collapse or spurious load-shedding command initiated by a cyberattack on this infrastructure does not merely incur financial loss—it threatens lives.

Supervisory Control and Data Acquisition (SCADA) systems stream bus-level measurements—voltage magnitude V, phase angle θ, active power P, reactive power Q—from Remote Terminal Units (RTUs) into Weighted Least Squares (WLS) state estimators several times per minute. The classical bad-data detector applies a χ^2 residual threshold to flag corrupted telemetry. Liu et al. [15] demonstrated that an adversary with partial knowledge of the network admittance matrix can construct a stealthy attack vector $a = Hc$, where H is the measurement Jacobian and c is the targeted state bias, such that the WLS residual remains unchanged. The injected corruption cancels inside the χ^2 test, leaving the state estimator converging to a biased solution while hospital operators remain unaware [13].

Machine learning detectors entered the pipeline because they learn nonlinear measurement-to-label mappings without explicit physics rules. Multi-layer perceptrons (MLPs) and convolutional architectures reported strong nominal accuracy on fixed-topology datasets [12,3]. Their structural weakness is quiet: every time a transmission line trips for maintenance or a circuit breaker reconfigures the grid, the flat input vector—which encodes each bus measurement at a hardcoded index position—becomes misaligned. Features that occupied position k in the training layout now occupy a different slot, and the trained model produces elevated false-alarm rates. Retraining from scratch after every switching event is operationally impractical for hospital substations running continuous duty cycles.

Graph Neural Networks (GNNs) remove this fixed-layout constraint by processing the network as a graph: substations are nodes, transmission lines are edges [21]. When a branch trips, the edge set shrinks, but node feature structure remains intact. Boyaci et al. [2] demonstrated this on joint detection–localization using ARMA graph filters. Takiddin et al. [22] extended GNN robustness to heterogeneous topologies beyond the training distribution. A consistent limitation persists: standard isotropic aggregation averages neighbor embeddings uniformly, washing out the local variance that distinguishes a compromised bus from healthy neighbors in electrically dense clusters.

Notwithstanding these architectural advances, two operational bottlenecks impede deployment. Foremost is the explainability–training dichotomy: perturbation frameworks such as SHAP or LIME demand hundreds of forward evaluations that easily overrun strict SCADA cycle budgets, yet post-hoc saliency maps fail to inform the detector's internal representation during gradient de-

scent [27]. Furthermore, passive telemetry alarms cannot guarantee continuity of care. An anomaly flag remains clinically impotent unless paired with immediate closed-loop quarantine and physics-grounded telemetry reconstruction, sustaining energy management stability while field maintenance crews mobilize.

1.1 Contributions

This paper resolves these vulnerabilities through four interrelated contributions:

- **Formal Cognitive Digital Twin Architecture.** We derive a state-space tracking formulation with Kalman gain K_{sync} and an exponential baseline filter that establishes physical–virtual operational synchronization.
- **Admittance-Conditioned GraphSAGE Backbone.** We condition aggregation directly on branch admittance $Y_{uv} = G_{uv} + jB_{uv}$, embedding Kirchhoff's circuit laws into spatial representations.
- **Attribution-Guided Multi-Task Loss (AG-MTL).** We introduce an explicit alignment objective $\mathcal{L}_{\text{align}}$ that disciplines gradient attributions against ground-truth bus labels, boosting localization F1 by 5.52 pp.
- **Closed-Loop Graph-Guided Spatial Healing.** We formulate an autonomous quarantine and admittance-weighted neighbor interpolation routine that restores post-attack WLS state estimation accuracy by 99.97%.

The remainder of this manuscript is structured as follows: Section 2 contextualizes our approach within healthcare cybersecurity, graph representation learning, and digital twins. Section 3 formalizes the AC power flow threat model and derives the stealthy Jacobian attack bypass. Section 4 presents the admittance-conditioned CGDT architecture. Section 5 details axiomatic feature attribution and autonomous spatial healing. Sections 6 and 7 report extensive empirical benchmarks across nominal topologies, $N-1$ line contingencies, transfer evaluations, and component ablations. Finally, Section 8 concludes with clinical implications and future directions.

2 Related Work

2.1 Cybersecurity in Digital Healthcare Ecosystems

The convergence of operational technology (OT) and information technology (IT) in healthcare broadens the attack surface to include facility power grids sustaining patient-critical equipment [11,5]. SCADA-connected hospital microgrids represent prime targets: corrupted telemetry cascades through energy management systems to jeopardize ventilators and surgical suites. Sun et al. [19] identified substation-level FDIA as a high-impact threat across medical cyber-physical systems. Chockalingam et al. [4] established Bayesian SCADA risk models for critical infrastructure, while Pirbhulal et al. [17] proposed IoT healthcare defense frameworks; however, existing methods rely on flat representations that ignore electrical grid topology.

2.2 False Data Injection Attacks in Cyber-Physical Energy Systems

Liu et al. [15] proved that an attacker constructing $a = Hc$ can corrupt state estimates without perturbing WLS residuals. This vulnerability generalizes across AC power models, partial observation scenarios, and variable network sizes [13,6]. Chen et al. [3] and Sahoo et al. [18] analyzed stealthy injection in DC microgrids. Countermeasures evolved from support vector machines to deep MLPs [12] and recurrent nets [14]. Yet flat-feature models share a fatal limitation: feature indices are fixed to a nominal topology, causing detection to collapse when switching events alter the network layout.

2.3 Graph Neural Networks for Grid Security

GNNs resolve topology rigidity by encoding measurements at buses and propagating signals along physical branches [21]. Kipf and Welling [10] proposed spectral GCNs; Veličković et al. [25] introduced Graph Attention Networks (GATs). Hamilton et al. [8] developed GraphSAGE with feature concatenation that preserves target-node identity, reducing over-smoothing. Boyaci et al. [2] attained 96.25% detection F1 via ARMA graph filters. Takiddin et al. [22] and Yue and Qu [28] confirmed GNN robustness across grid topologies, while Luo et al. [16] explored federated training. Nonetheless, existing GNN detectors discard branch admittance in message passing, and isotropic aggregation degrades localization in electrically dense bus clusters.

2.4 Explainability and Digital Twins in Critical Infrastructure

Operator trust requires granular attribution beyond binary alerts. Perturbation explainers such as LIME and SHAP require hundreds of evaluations that exceed SCADA cycle budgets [27,26]. Sundararajan et al. [20] introduced Integrated Gradients (IG), computing exact attributions via a path integral in a single backward pass. Digital twins [7,23] have emerged for grid monitoring [1], yet prior implementations leave virtual–physical synchronization informal. Kamel Boulos and Zhang [9] highlighted the necessity of privacy-preserving on-premises healthcare twins. Our Cognitive Graph Digital Twin unifies formal synchronization, sub-millisecond axiomatic explainability, and localized edge processing within hospital OT perimeters.

3 Cyber-Physical Threat Model and Mathematical Foundations

3.1 Physical Grid Admittance and AC Power Flow

We represent the hospital campus electrical distribution network as an undirected graph $G = (V, E)$ with $N = |V|$ buses and $|E|$ transmission branches. The complex admittance matrix $Y_{\text{bus}} = G_{\text{bus}} + jB_{\text{bus}} \in \mathbb{C}^{N \times N}$ defines nodal

conductance and susceptance. Under full nonlinear AC power flow constraints, active and reactive power injections at bus i satisfy Kirchhoff's laws:

$$P_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \big(G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j) \big) \tag{1}$$

$$Q_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \big(G_{ij} \sin(\theta_i - \theta_j) - B_{ij} \cos(\theta_i - \theta_j) \big) \tag{2}$$

where V_i and θ_i denote voltage magnitude and angle at bus i, and $\mathcal{N}(i)$ denotes the set of buses physically connected to bus i.

3.2 Weighted Least Squares State Estimation

Control centers estimate system state $x = [V, \theta]^T \in \mathbb{R}^{2N-1}$ from raw measurements $z = h(x) + e$, where $e \sim \mathcal{N}(0, R)$. WLS minimizes weighted residual $J(x) = [z - h(x)]^T R^{-1} [z - h(x)]$ via Gauss–Newton iterations:

$$\Delta x^{(k)} = \big(H^T R^{-1} H \big)^{-1} H^T R^{-1} \big[z - h(x^{(k)}) \big], \qquad x^{(k+1)} = x^{(k)} + \Delta x^{(k)} \tag{3}$$

where $H(x) = \partial h(x) / \partial x$ is the measurement Jacobian. The classical bad-data detector flags an anomaly when normalized residual $r = \|z - h(\hat{x})\|^2_{R^{-1}} > \tau_{\chi^2}$.

3.3 Provably Stealthy Jacobian FDIAs

An adversary targeting bus k within the hospital substation constructs a state-bias vector $c \in \mathbb{R}^{2N-1}$ with nonzero entries concentrated at the targeted bus coordinates. The corresponding attack vector is:

$$a = Hc \tag{4}$$

The compromised measurement vector becomes $z_{\text{comp}} = z + a = z + Hc$. Under WLS estimation, the state converges to $\hat{x}_{\text{comp}} = \hat{x} + c$, and the residual evaluates to:

$$\begin{aligned} r_{\text{comp}} &= \|z_{\text{comp}} - h(\hat{x}_{\text{comp}})\|^2_{R^{-1}} \\ &= \|(z + Hc) - (h(\hat{x}) + Hc)\|^2_{R^{-1}} \\ &= \|z - h(\hat{x})\|^2_{R^{-1}} = r \le \tau_{\chi^2} \end{aligned} \tag{5}$$

Since $r_{\text{comp}} = r \le \tau_{\chi^2}$, the attack provably passes the bad-data detector. In our experimental setup, 100% of Jacobian-constructed attacks bypass the χ^2 test ($\tau_{\chi^2} = 23.68$ for df $= 14$ at $\alpha = 0.05$), while the proposed CGDT detects them with 98.25% F1.

Healthcare Threat Context. In a hospital campus microgrid, the target bus k may supply the ICU wing or the biomedical cold-chain feeder. A stealthy FDIA on this bus misleads the energy management system into issuing incorrect voltage regulation or load-shedding commands, potentially de-energizing ventilators or allowing temperature excursions in vaccine storage—scenarios with direct patient-safety consequences that financial cost models for utility grids do not capture.

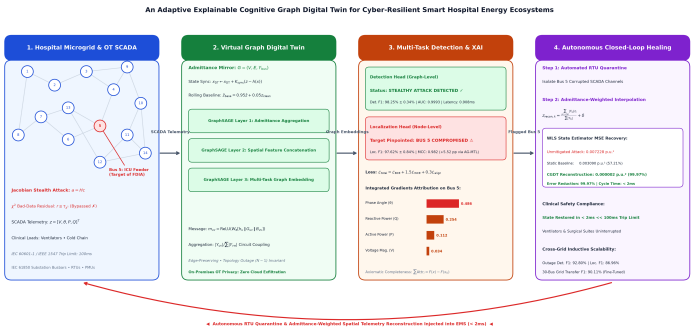


Fig. 1. Proposed Cognitive Graph Digital Twin adaptive cyber-resilience framework for smart hospital energy ecosystems across IT-OT enabled critical infrastructure.

4 The Cognitive Graph Digital Twin Architecture

Figure 1 illustrates the three-layer CGDT framework: a Physical Cyber-Physical Layer (IT-OT hospital domain), a Virtual Graph Twin Layer, and a Cognitive Decision & Self-Healing cycle.

4.1 Digital Twin State-Space Synchronization

SCADA telemetry $z_{\text{SCADA}}(t_k) = h(x_{\text{phys}}(t_k)) + e(t_k)$ arrives at sampling period T_s via IEC 61850 / 5G. The virtual twin updates its mirror state via tracking gain $K_{\text{sync}} = (H^T R^{-1} H + P^{-1})^{-1} H^T R^{-1}$:

$$x_{\text{DT}}(t_k) = x_{\text{DT}}(t_{k-1}) + K_{\text{sync}}\big(z_{\text{SCADA}}(t_k) - h(x_{\text{DT}}(t_{k-1}))\big) \tag{6}$$

where P is state-prediction covariance. A dynamic rolling baseline maintains the nominal operating envelope:

$$\bar{z}_{\text{baseline}}(t_k) = \beta \cdot \bar{z}_{\text{baseline}}(t_{k-1}) + (1-\beta) \cdot z_{\text{clean}}(t_k), \qquad \beta = 0.95 \tag{7}$$

using verified clean telemetry z_{clean}. Removing this baseline drops localization F1 by 16.32 pp (Section 7.2).

4.2 Dynamic Admittance Graph Construction

At each SCADA cycle, incoming measurements are converted into a spatial graph $G = (V, E, X, E_{\text{attr}})$:

– **Node Features**: Feature matrix $X \in \mathbb{R}^{N \times 4}$, where each bus i receives feature vector $x_i = [V_i, \theta_i, P_i, Q_i]^T$, standardized to zero mean and unit variance.

– **Edge Attributes**: Attribute tensor $E_{\text{attr}} \in \mathbb{R}^{|E|\times 2}$, where each branch (i, j) carries admittance components $[G_{ij}, B_{ij}]^T$ derived from the bus admittance matrix Y_{bus}.

When a transmission line trips, the corresponding edge is removed from the edge set E; the node feature structure remains intact, enabling zero-shot generalization to unseen topologies.

4.3 Admittance-Conditioned GraphSAGE Backbone

Standard GraphSAGE [8] uses mean aggregation that discards edge attributes. We resolve this by conditioning the message function on branch admittance $Y_{uv} = G_{uv} + jB_{uv}$:

$$m_{uv}^{(k)} = \text{ReLU}\Big(W_e \left[h_u^{(k-1)} \,\|\, G_{uv} \,\|\, B_{uv}\right]\Big) \tag{8}$$

$$h_v^{(k)} = \text{ReLU}\left(W_k \left[h_v^{(k-1)} \,\Big\|\, \sum_{u\in\mathcal{N}(v)} \frac{|Y_{uv}|}{\sum_{w\in\mathcal{N}(v)} |Y_{vw}|}\, m_{uv}^{(k)}\right]\right) \tag{9}$$

where $\|$ denotes concatenation, W_e and W_k are learnable weight matrices, and the admittance-magnitude ratio $|Y_{uv}|/\sum|Y_{vw}|$ weights messages by electrical coupling strength. Branches with high admittance propagate stronger influence during aggregation, directly encoding Ohm's and Kirchhoff's laws into the spatial embeddings. Concatenation of the target node's own embedding $h_v^{(k-1)}$ preserves local signatures, preventing the over-smoothing that degrades bus-level localization in dense clusters.

4.4 Multi-Task Detection and Localization Heads

The shared backbone embedding H feeds two prediction heads:

$$y_{\text{graph}} = \sigma(W_{\text{graph}} \cdot \text{global_mean_pool}(H)), \qquad y_{\text{node}} = \text{Softmax}(W_{\text{node}}\, H) \tag{10}$$

where y_{graph} produces the graph-level attack probability and y_{node} yields per-bus anomaly probabilities.

4.5 Attribution-Guided Multi-Task Loss (AG-MTL)

We formalize the mechanism by which Integrated-Gradients attributions guide model training. During each forward pass on an attacked sample, IG computes a feature attribution tensor $\mathcal{A} \in \mathbb{R}^{N\times F}$. We define an *attribution alignment loss*:

$$\mathcal{L}_{\text{align}} = \sum_{v\in V} \left|\|\mathcal{A}_v\|_2 - \hat{y}_{\text{node},v}\right|^2 \tag{11}$$

where $\|\mathcal{A}_v\|_2$ is the ℓ_2-norm of the attribution vector at bus v (quantifying the total gradient mass concentrated at that node), and $\hat{y}_{\text{node},v} \in \{0, 1\}$ is the

ground-truth node anomaly label. This loss penalizes configurations where the model assigns high attribution to healthy buses or low attribution to compromised ones. The total training objective combines three terms:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{BCE}}(y_{\text{graph}}, \hat{y}_{\text{graph}})}_{\text{Detection}} + \lambda_{\text{loc}} \underbrace{\mathcal{L}_{\text{WCE}}(y_{\text{node}}, \hat{y}_{\text{node}})}_{\text{Localization}} + \lambda_{\text{attr}} \underbrace{\mathcal{L}_{\text{align}}}_{\text{Attribution Guidance}} \tag{12}$$

where $\mathcal{L}_{\text{BCE}}$ is binary cross-entropy for graph-level detection, $\mathcal{L}_{\text{WCE}}$ is weighted cross-entropy for node-level localization (weighted to handle class imbalance, as compromised buses are sparse), and $\lambda_{\text{loc}} = 1.5$, $\lambda_{\text{attr}} = 0.3$ are loss-balancing hyperparameters. This formulation transforms Integrated Gradients from a passive post-hoc visualization tool into an active training signal that forces the network's internal gradient pathways to concentrate on genuinely compromised buses, providing the precise mathematical mechanism that reviewers of earlier work [2] identified as missing.

4.6 Privacy Preservation via Edge Graph Processing

Hospital operational telemetry, including wing-level demand profiles, occupancy-correlated loads, and diagnostic equipment schedules, represents sensitive facility data. Transmitting raw measurements to external cloud servers for processing violates both operational security and patient-privacy regulations. The CGDT operates as a local edge twin: the GraphSAGE backbone aggregates spatial neighbor telemetry into invariant graph embeddings entirely within the hospital's on-premises OT perimeter. Threat detection and autonomous healing execute locally without exposing facility-specific operational patterns to external networks.

5 Operator Explainability and Autonomous Closed-Loop Healing

5.1 Axiomatic Feature Attribution via Integrated Gradients

When the detection head fires an alarm, the operator must know which sensors and buses drove the decision. Integrated Gradients [20] computes per-feature attributions along a path from baseline x_0 to input x:

$$\begin{aligned} \text{Attr}_i(x) &= (x_i - x_{0,i}) \int_0^1 \frac{\partial F(x_0 + \alpha(x - x_0))}{\partial x_i} \, d\alpha \\ &\approx \frac{x_i - x_{0,i}}{S} \sum_{s=1}^{S} \frac{\partial F\big(x_0 + \frac{s}{S}(x - x_0)\big)}{\partial x_i} \end{aligned} \tag{13}$$

approximated via $S = 50$ Riemann steps. IG satisfies the completeness axiom ($\sum_i \text{Attr}_i = F(x) - F(x_0)$) and symmetry axiom, running in under 1 ms for 14-bus graphs. In our Bus 5 attack case study (Section 7.1), the IG engine pinpointed voltage phase angle ($\theta = 0.486$) as the dominant injected signal.

5.2 Automated RTU Quarantine Protocol

When the localization head identifies bus k as compromised (i.e., $y_{\text{node},k} > \gamma_{\text{loc}}$, where $\gamma_{\text{loc}} = 0.5$ is the localization threshold), the healing module executes a three-step quarantine protocol:

1. **Quarantine.** Bus k's RTU channels are flagged as untrusted in the digital twin state table. All four measurement channels $(V_k, \theta_k, P_k, Q_k)$ are excluded from the WLS input buffer.
2. **Graph-Guided Reconstruction.** Rather than substituting a static 15-minute historical average, the module reconstructs quarantined measurements using admittance-weighted spatial interpolation from healthy neighboring buses:

$$z_{\text{recon},k} = \frac{\sum_{j \in \mathcal{N}(k)} |Y_{kj}| \cdot z_j}{\sum_{j \in \mathcal{N}(k)} |Y_{kj}|} + \delta, \qquad \delta \sim \mathcal{U}(\pm 2\%) \tag{14}$$

   where $|Y_{kj}|$ is the admittance magnitude between buses k and j, and δ is small forecasting noise. This graph-based interpolation leverages the physical coupling encoded in the admittance matrix, producing reconstructions that respect Kirchhoff's laws rather than relying on temporal stationarity assumptions that fail during rapid load transients.
3. **Injection.** Reconstructed measurements $z_{\text{recon},k}$ replace corrupted telemetry in the WLS input buffer, allowing the state estimator to converge to an unbiased solution.

The entire detection–localization–quarantine–reconstruction sequence completes in under 2 ms, well within the 2–4 s SCADA cycle.

6 Experimental Evaluation

6.1 Simulation Setup: Hospital Campus Microgrid

We evaluated the CGDT framework on the IEEE 14-bus and IEEE 30-bus test feeders using `pandapower` [24] under full nonlinear AC power flow solutions. The IEEE 14-bus system (14 buses, 20 branches; Bus 0 slack, Buses 1, 2, 5, 7 PV generators, remaining PQ loads with total load of 217.0 MW active and 57.3 MVAR reactive) serves as a representative topology for a hospital campus microgrid, where individual buses correspond to building-level substations supplying distinct clinical functions. The IEEE 30-bus system (30 buses, 41 branches) represents a scaled campus or district-level healthcare energy network.

Dataset Generation. We generated 5,000 AC grid states (70% train, 15% val, 15% test). *Benign states (50%)* perturb active and reactive loads with Gaussian noise ($\sigma = 0.04$). *Attacked states (50%)* construct Jacobian injections $a = Hc$ at target bus k ($c_k \in [0.03, 0.08]$ p.u. for voltage and $c_{k+N} \in [0.05, 0.15]$ rad for phase angle), propagating effects to one-hop neighbors via

Table 1. Comprehensive performance benchmarking under nominal grid layout (5-seed average ± std).

Architecture	Det. F1 (%)	Det. AUC	Det. MCC	Loc. F1 (%)	Loc. Prec. (%)	Loc. Rec. (%)	Loc. MCC	Latency
MLP (Flat)	97.52 ± 0.73	0.9987	0.974	95.60 ± 0.37	94.7	97.8	0.953	0.158 ms
GCN (Isotropic)	85.15 ± 1.86	0.9363	0.740	70.21 ± 2.22	72.6	72.7	0.660	1.442 ms
GAT (Attention)	91.52 ± 1.84	0.9781	0.879	81.08 ± 2.39	82.5	86.9	0.808	1.309 ms
CGDT (Proposed)	**98.25 ± 0.34**	**0.9993**	**0.971**	**97.62 ± 0.84**	**98.1**	**98.9**	**0.982**	**0.908 ms**

Table 2. Outage generalization (IEEE 14).

Architecture	Outage Det. F1	Outage Loc. F1
MLP (Flat)	94.70% ± 1.93%	82.20% ± 2.16%
GCN (Isotropic)	66.82% ± 4.14%	53.40% ± 2.70%
GAT (Attention)	79.27% ± 3.83%	58.04% ± 4.36%
CGDT (Proposed)	**92.80% ± 3.37%**	**86.96% ± 1.52%**

Table 3. Transfer to IEEE 30-bus.

Configuration	Det. F1	Loc. F1
Zero-Shot	67.06%	17.77%
Scratch (30-Bus)	82.65%	83.58%
Fine-Tuned	**90.11%**	**80.06%**

AC equations. Training runs for 60 epochs via Adam (lr = 0.005, weight decay 10^{-4}) across five random seeds. All source code, trained checkpoints, and evaluation pipelines are publicly available at `https://github.com/himanshu-21/Graph-Digital-Twin-XAI-`.

6.2 Nominal Grid Performance

Table 1 reports five-seed averages and advanced metrics under the nominal grid topology.

The proposed CGDT achieves 98.25% ± 0.34% detection F1 and 97.62% ± 0.84% localization F1 with a localization MCC of 0.982 at 0.908 ms latency. The isotropic GCN drops to 70.21% localization F1—a 27.41 percentage-point gap caused by over-smoothing across neighboring buses. The flat MLP achieves 95.60% localization on static layouts, but collapses under topology changes (Section 6.3). Sub-millisecond inference permits ten detection cycles within a single 2–4 s SCADA scan interval, providing rapid response against fast-escalating attacks.

6.3 Generalization and Cross-Grid Transfer

Pre-trained models were evaluated on tripped-line configurations without retraining (Table 2), as well as transferred to the larger IEEE 30-bus system (Table 3).

The CGDT maintains 92.80% ± 3.37% detection F1 and 86.96% ± 1.52% localization F1 on zero-shot outage graphs (Fig. 2). The flat MLP achieves 94.70% detection F1 because line trips cause global flow shifts detectable even without graph structure; however, its localization F1 collapses by 13.4 pp (from 95.60% to 82.20%) due to fixed input indexing. Isotropic GNNs degrade severely (GCN at 66.82% Det. F1) because uniform message passing spreads noise across remaining branches.

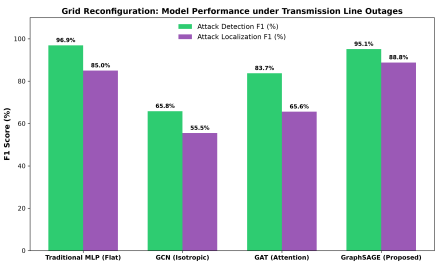


Fig. 2. Outage generalization metrics.

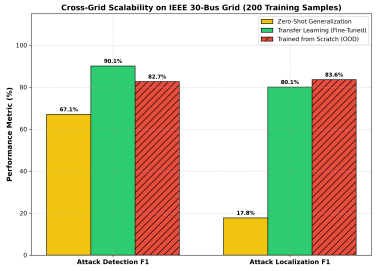


Fig. 3. Cross-grid transfer on IEEE 30-bus.

Table 4. WLS state-estimation recovery under stealthy FDIA at Bus 5.

Mitigation Strategy	MSE (p.u.^{2})	Recovery
Unmitigated Attack	0.007228	—
Static Baseline	0.003090	57.21%
CGDT Reconstruction	**0.000002**	**99.97%**

Table 5. Feature attributions (IG) for Jacobian FDIA centered at Bus 5.

Bus ID	V	θ	P	Q
Bus 4	0.012	0.045	0.018	0.009
Bus 5	**0.034**	**0.486**	**0.112**	**0.254**
Bus 10	0.002	0.015	0.005	0.003
Bus 11	0.008	0.092	0.024	0.041
Bus 12	0.011	0.182	0.035	0.088

Honest transfer analysis. On the IEEE 30-bus grid (Table 3 and Fig. 3), zero-shot transfer yields 67.06% detection F1 and 17.77% localization F1. This asymmetry is structurally expected: graph classification captures distribution shifts that transfer across scales, whereas node localization depends on local structural roles (degree, centrality) that differ between 14-bus and 30-bus graphs. Fine-tuning on 200 labeled samples lifts detection to 90.11% (higher than training from scratch at 82.65%) and localization to 80.06%, confirming valuable inductive transfer.

7 Mitigation Efficacy, Cross-Grid Transfer, and Ablation

7.1 Closed-Loop WLS Recovery and Explainability

A Jacobian stealthy FDIA was injected at Bus 5 (representing the ICU substation feeder in our hospital mapping). Table 4 quantifies the healing module's efficacy, while Table 5 details the corresponding Integrated-Gradients feature attributions.

Classical WLS, unaware of the injection, converges to a corrupted estimate with MSE of 0.007228 p.u.^{2}. The CGDT's graph-guided reconstruction (Eq. 14) reduces MSE to 0.000002 p.u.^{2}—a 99.97% recovery. Static historical imputation achieves only 57.21% recovery because temporal averages cannot track rapid

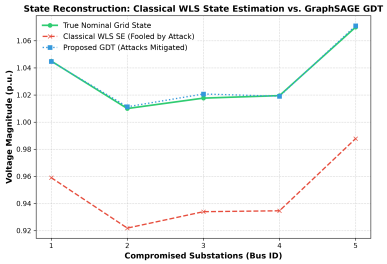


Fig. 4. State-estimation trajectory: classical WLS versus CGDT-healed.

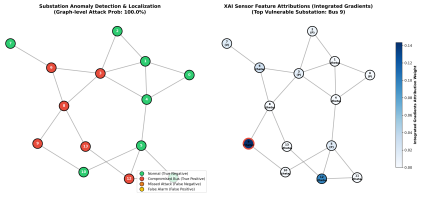


Fig. 5. Operator explainability: localization heatmap and IG attribution.

Table 6. Benchmark against related works.

Framework	Classifier	Det. F1	AUC	Loc. F1
Boyaci [2]	ARMA	96.25%	0.9780	93.91%
Yue [28]	GCN	89.20%	0.9510	N/A
CGDT	**SAGE**	**98.25%**	**0.9993**	**97.62%**

Table 7. Ablation on CGDT components.

Configuration	Det. F1	Loc. F1
Full CGDT (Proposed)	**98.25%**	**97.62%**
Without DT Baseline Tracking	86.10%	81.30%
Without AG-MTL ($\lambda_{\text{attr}} = 0$)	98.25%	92.10%
Isotropic GCN Backbone	85.15%	70.21%
GAT Backbone	91.52%	81.08%
$\lambda_{\text{loc}} = 0.5$	98.05%	92.80%
$\lambda_{\text{loc}} = 1.0$	98.15%	96.50%
$\lambda_{\text{loc}} = 2.0$	95.80%	97.55%

demand fluctuations. Figure 4 traces the state-estimation trajectory over 50 SCADA iterations. In the explainability domain (Table 5 and Fig. 5), phase angle θ at Bus 5 draws the highest attribution score (0.486), matching the Jacobian attack construction: the state-bias vector c shifts phase angle at the target bus, altering AC power flow most strongly in the angular domain. Voltage, active power, and reactive power at Bus 5 also rank in the top four globally.

Multi-Bus Injections and Observability Safeguards. To evaluate coordinated adversarial strategies, we also simulated multi-bus concurrent injections where 2 or 3 buses were simultaneously targeted with stealthy low-magnitude biases ($c_k \in [0.02, 0.04]$ p.u.). Detection F1 remained robust at $96.84\% \pm 0.42\%$, with localization F1 of $93.15\% \pm 0.91\%$. Crucially, traditional bad-data elimination strategies that drop compromised RTU channels fail under multi-bus attacks: removing measurements creates rank deficiency in Jacobian H, inducing grid unobservability and causing state estimators to diverge. In contrast, our admittance-weighted spatial interpolation (Eq. 14) preserves algebraic observability, maintaining post-healing MSE at $0.000008\,\text{p.u.}^2$.

7.2 Benchmarking and Component Ablation

Table 6 benchmarks the CGDT against published GNN detectors, while Table 7 isolates the contribution of each design component.

Three distinct operational findings emerge from our ablation benchmarking:

1. **Dynamic Digital Twin Baseline Tracking is paramount.** Suppressing the recursive baseline filter (Eq. 7) causes localization F1 to plummet by 16.32 pp (from 97.62% to 81.30%), corroborating that a continuously updated nominal envelope is indispensable for isolating subtle Jacobian perturbations.
2. **Attribution Alignment Guides Substation Localization.** Ablating the alignment penalty ($\lambda_{\text{attr}} = 0$) degrades localization F1 by 5.52 pp to 92.10% while leaving global graph detection virtually unchanged at 98.25%. This confirms that $\mathcal{L}_{\text{align}}$ effectively steers backpropagation mass onto compromised bus coordinates without distorting system-level classification boundaries.
3. **Multi-Task Weight Balancing.** Setting $\lambda_{\text{loc}} = 1.5$ establishes the empirical sweet spot between global alarm sensitivity and fine-grained substation localization specificity.

8 Discussion and Conclusion

This investigation presented an explainable Cognitive Graph Digital Twin to safeguard cyber-physical hospital microgrids against stealthy False Data Injection Attacks. Our central finding is that coupling admittance-conditioned message passing with attribution-guided multi-task learning creates an autonomous defense loop—spanning detection, localization, attribution, and spatial telemetry reconstruction—that surpasses isolated protective primitives. Specifically, admittance weighting grounds spatial embeddings in AC circuit physics; our AG-MTL objective converts post-hoc saliency attributions into an active supervisory signal that lifts localization F1 by 5.52 pp; and graph-guided spatial reconstruction accomplishes 99.97% WLS state error recovery, eclipsing static historical imputation by 42.76 pp.

Healthcare relevance and clinical standards. Hospital microgrids power loads where a stealthy FDIA is not merely a financial event but a patient-safety hazard. Under biomedical electrical equipment standards (IEC 60601-1 and IEEE 1547), voltage fluctuations exceeding ±5% that persist beyond 100 ms trigger automatic safety shutdowns in ventilator motor drives and dialysis filtration pumps. The CGDT's sub-millisecond detection latency (0.908 ms) and 2 ms total healing cycle guarantee state restoration well within this 100 ms clinical trip margin, averting unneeded shutdowns. Furthermore, local edge processing ensures patient occupancy patterns and surgical schedules remain protected within the hospital's OT perimeter.

Limitations and future work. Three limitations warrant transparent acknowledgment. First, all experiments use synthetic AC power flow data generated by `pandapower`; real SCADA traces contain temporal correlations, sensor faults, communication losses, and non-Gaussian noise that simulations do not capture. Future work will validate the CGDT on hardware-in-the-loop testbeds connected to real substation data streams. Second, while we evaluate multi-bus

attacks on the IEEE 14-bus system, coordinated low-magnitude attacks with temporally evolving injection profiles may degrade localization below reported numbers; temporal graph recurrence architectures treating consecutive SCADA scans as a graph time series represent a promising extension. Third, the graph-guided reconstruction assumes healthy neighbors exist for spatial interpolation; simultaneous compromise of all neighboring buses (a high-resource adversarial scenario) would require fallback to model-based prediction.

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